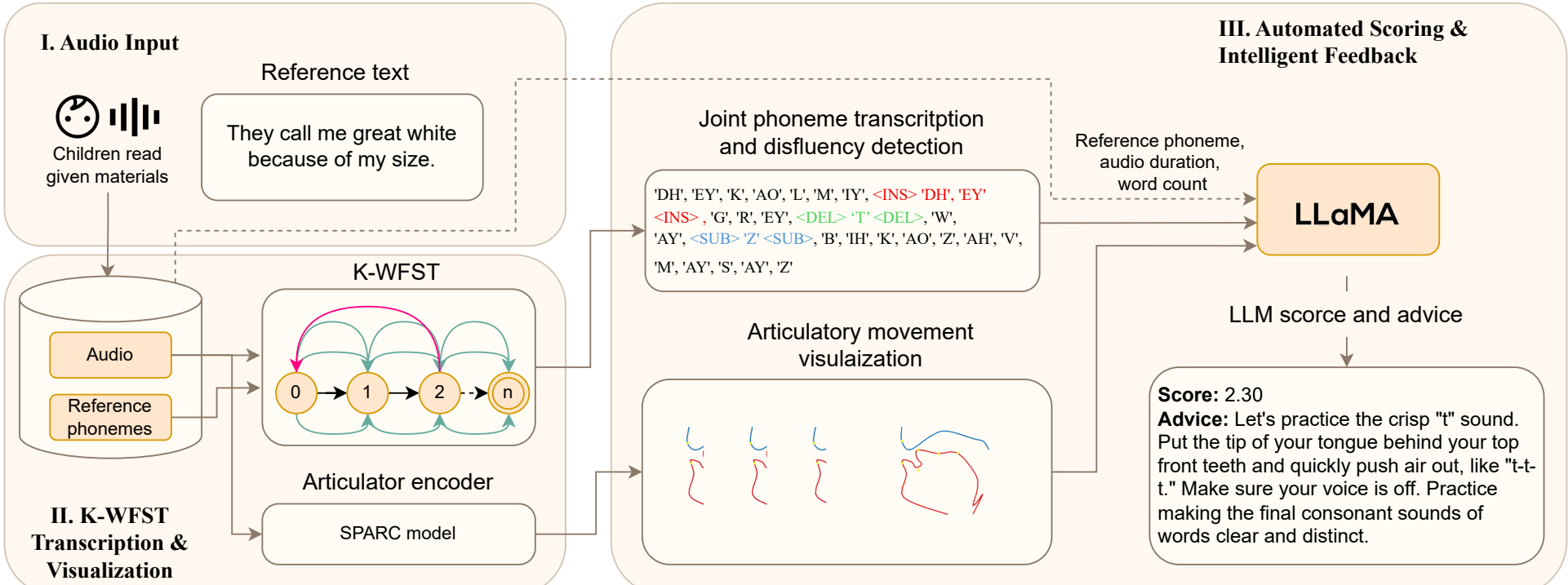
I. Audio Input
Children read given materials
Reference text
They call me great white because of my size.
Audio
Reference phonemes
K-WFST
0
1
2
n
Articulator encoder
SPARC model
II. K-WFST Transcription & Visualization
Joint phoneme transcritption and disfluency detection
'DH', 'EY', 'K', 'AO', 'L', 'M', 'IY', <INS> 'DH', 'EY' <INS> , 'G', 'R', 'EY', <DEL> 'T' <DEL>, 'W', 'AY', <SUB> 'Z' <SUB>, 'B', 'IH', 'K', 'AO', 'Z', 'AH', 'V', 'M', 'AY', 'S', 'AY', 'Z'
Articulatory movement visulaization
Reference phoneme, audio duration, word count
LLaMA
LLM score and advice
Score: 2.30
Advice: Let's practice the crisp "t" sound. Put the tip of your tongue behind your top front teeth and quickly push air out, like "t-t-t." Make sure your voice is off. Practice making the final consonant sounds of words clear and distinct.
III. Automated Scoring & Intelligent Feedback